

Common Effect Sizes and Conversions

Effect sizes are standardized metrics that describe the strength and direction of relationships, differences, or associations between variables. This handout provides an overview of the most common effect size metrics used in applied research and meta-analysis, along with conversions between them.

1. Common Effect Sizes

Effect Size	Formula / Definition	Type of Data	Interpretation
Cohen's d	$(M_1 - M_2) / SD_{pooled}$	Difference between two means	0.2 small, 0.5 medium, 0.8 large
Hedges' g	$J \times d$ (bias-corrected d)	Difference between two means	Adjusts d for small samples
Glass's Δ	$(M_1 - M_2) / SD_{control}$	Difference between two means	Uses control group SD only
r (correlation)	$Cov(X,Y) / (SD_x \times SD_y)$	Continuous variables	Strength of association (-1 to +1)
η^2 (eta squared)	$SS_{between} / SS_{total}$	ANOVA designs	Proportion of variance explained
ω^2 (omega squared)	$(SS_{between} - df \times MS_{within}) / (SS_{total} + MS_{within})$	ANOVA designs	Less biased η^2
Odds Ratio (OR)	$(a/b) / (c/d)$	Binary outcomes	Effect of exposure or treatment
Risk Ratio (RR)	$(a/(a+b)) / (c/(c+d))$	Binary outcomes	Relative likelihood of event
β (standardized regression)	$b \times (SD_x / SD_y)$	Continuous predictors	Predictor influence in SD units

2. Common Conversions Between Effect Sizes

From	To	Conversion Formula
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d	r	$r = d / \sqrt{(d^2 + 4)}$
r	d	$d = 2r / \sqrt{(1 - r^2)}$
t (two groups)	d	$d = t \times \sqrt{(1/n_1 + 1/n_2)}$
F (1, df ₂)	d	$d = 2 \times \sqrt{(F / df_2)}$
η^2	d	$d = 2\sqrt{(\eta^2 / (1 - \eta^2))}$
OR	d	$d = \ln(OR) \times \sqrt{3} / \pi$
r	R ²	$R^2 = r^2$
β	r	$r \approx \beta$ (in simple regression)

3. Selected References

Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Hillsdale, NJ: Lawrence Erlbaum Associates.

Hedges, L. V. (1981). *Distribution theory for Glass's estimator of effect size and related estimators.* Journal of Educational Statistics, 6(2), 107–128.

Rosenthal, R. (1991). Meta-analytic procedures for social research (Rev. ed.). Newbury Park, CA: Sage.

Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2009). *Introduction to Meta-Analysis.* Chichester, UK: Wiley.

Lipsey, M. W., & Wilson, D. B. (2001). *Practical Meta-Analysis.* Thousand Oaks, CA: Sage.